

Crosswork Assurance & IPM

SRv6 & IPM Roadshow

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IP is back and better than ever



Build
anything

Simplified, scalable,
and versatile networks
that are self-sufficient

Self-sufficiency is standard



End-to-end policy

- From Host to Internet through DC, Access, Metro, Core, Cloud
- No protocol conversion or gateways at domain boundaries



Any service, without any shim

- VPN, Slicing, Traffic Engineering, Green Routing, FRR, NFV, AI Flow Placement



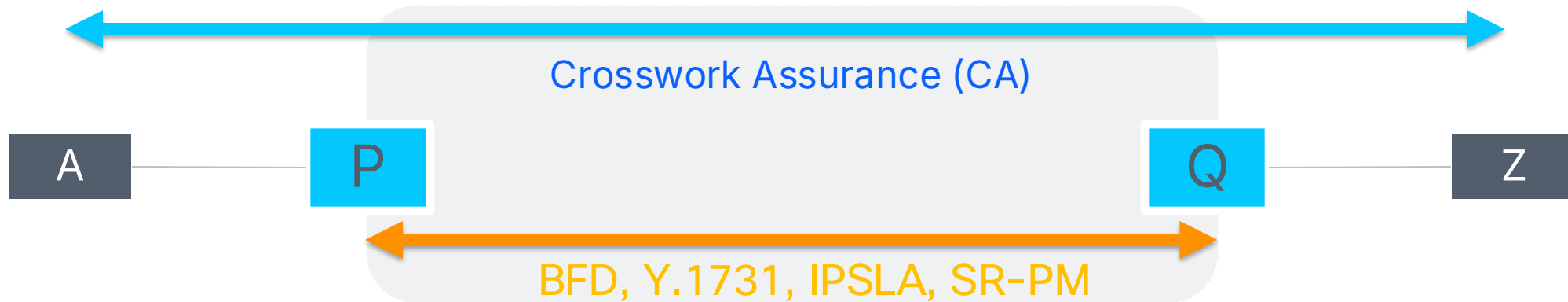
Better scale, reliability, cost, and seamless deployment in Brownfield



This is powerful — but ...
how do we prove it?

How do we know every path, every service, every millisecond is performing as intended?

Traditional Overlay Network Probing



- Active probing from A to Z provides the ability to measure SLAs and understand the end to end service performance for SLA backed services
 - Accurate, granular, directional
- Active probing from P to Q provides the ability to measure Liveness, Loss, Latency and understand the transport underlay performance
 - Embedded, flexible, prevalent

Traditional Overlay Network Probing Falls Short (P-Q)

- **Scalability Limits**

- Can only handle a few thousand probes per second.
- Fails to cover all ECMP paths—up to 10,000x less scalable than needed.
- Less than 1% path coverage—insufficient for AI-scale data centers.
- Example: 500 edge nodes * 16 ECMP paths * 1 probe every millisecond

- **Sub-optimal Latency Metrics***

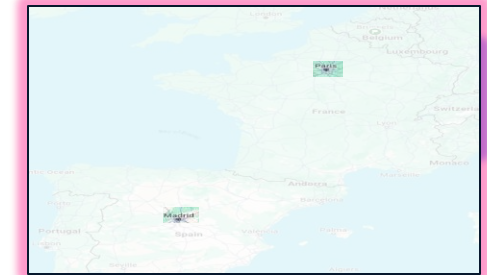
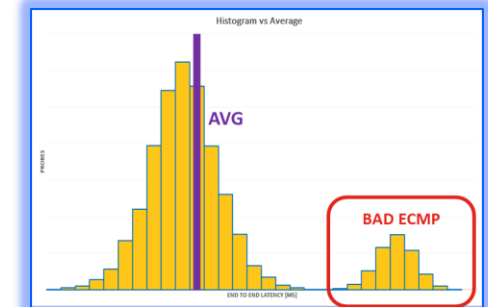
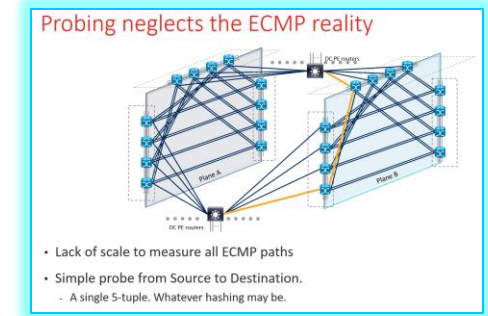
- Min/Max/Average values don't capture the full user experience.
- Example: 1 bad path out of 8 =12.5% of the clients impacted. Average hides the issue.

- **Path Asymmetry***

- Upstream ≠ Downstream: Loss and liveness issues are hard to localize in 2-way methods.
- No clarity on where the problem lies—source or destination.

- **Lack of Visibility**

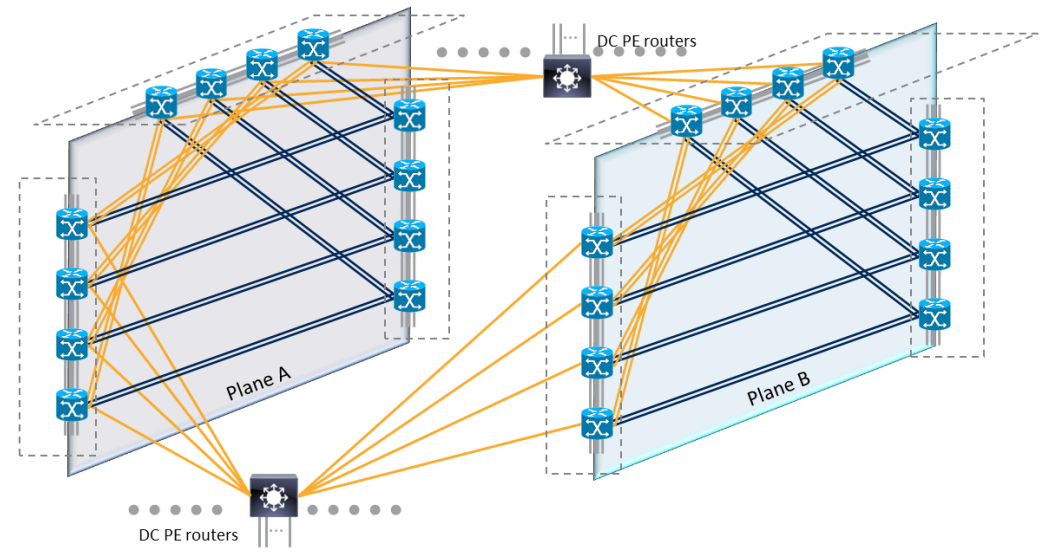
- The underlay network is a black box.
- No insight into how traffic flows—makes SLA validation unreliable.
- Example: How do we know if Paris to Madrid latency is acceptable?



*Applies mostly to transport probing solutions

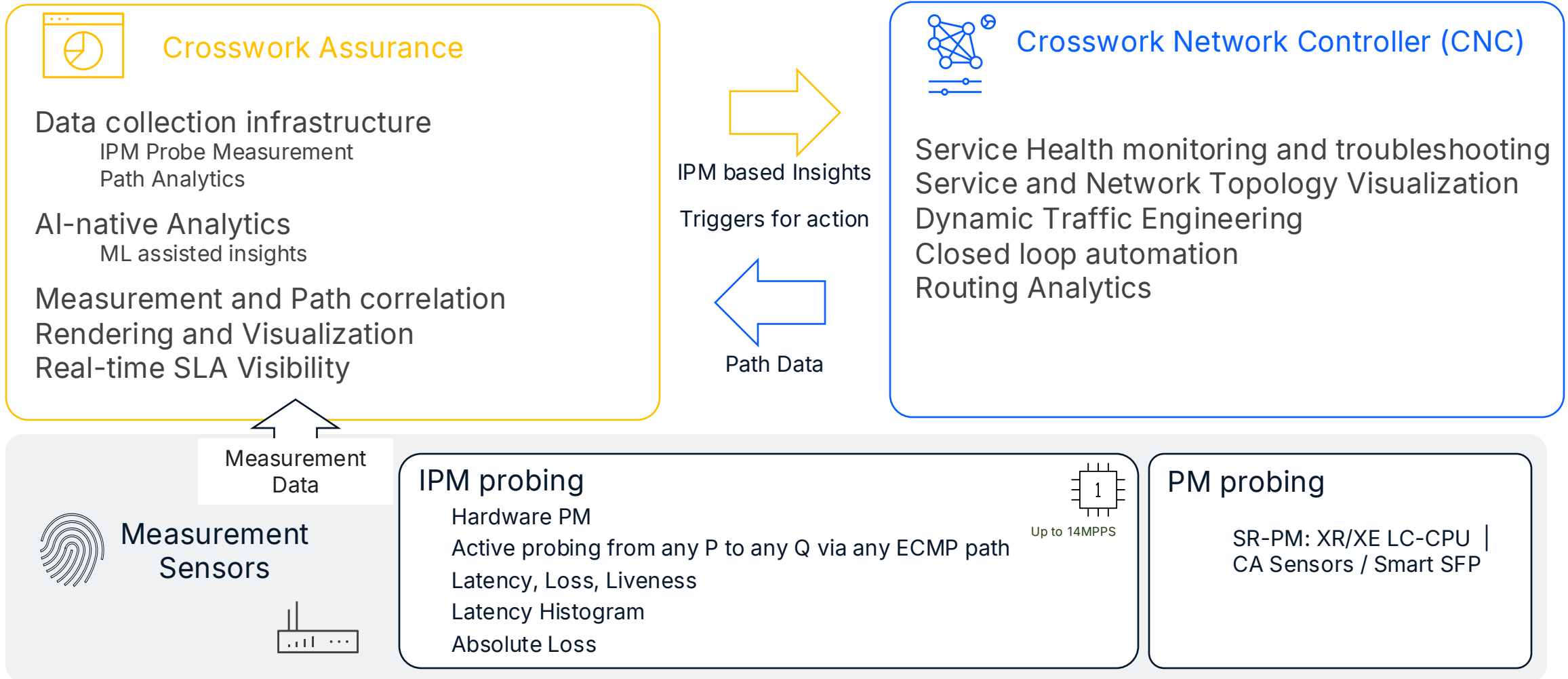
We need a solution that:

- Discovers and monitors **all ECMP paths**
- Delivers **enough PPS** to measure all ECMP paths
- Reports **accurately** across ECMP paths
- Provides continuous **routing monitoring**
- Unleashes the power of **Analytics**
 - Correlates probe measurement and routing path data

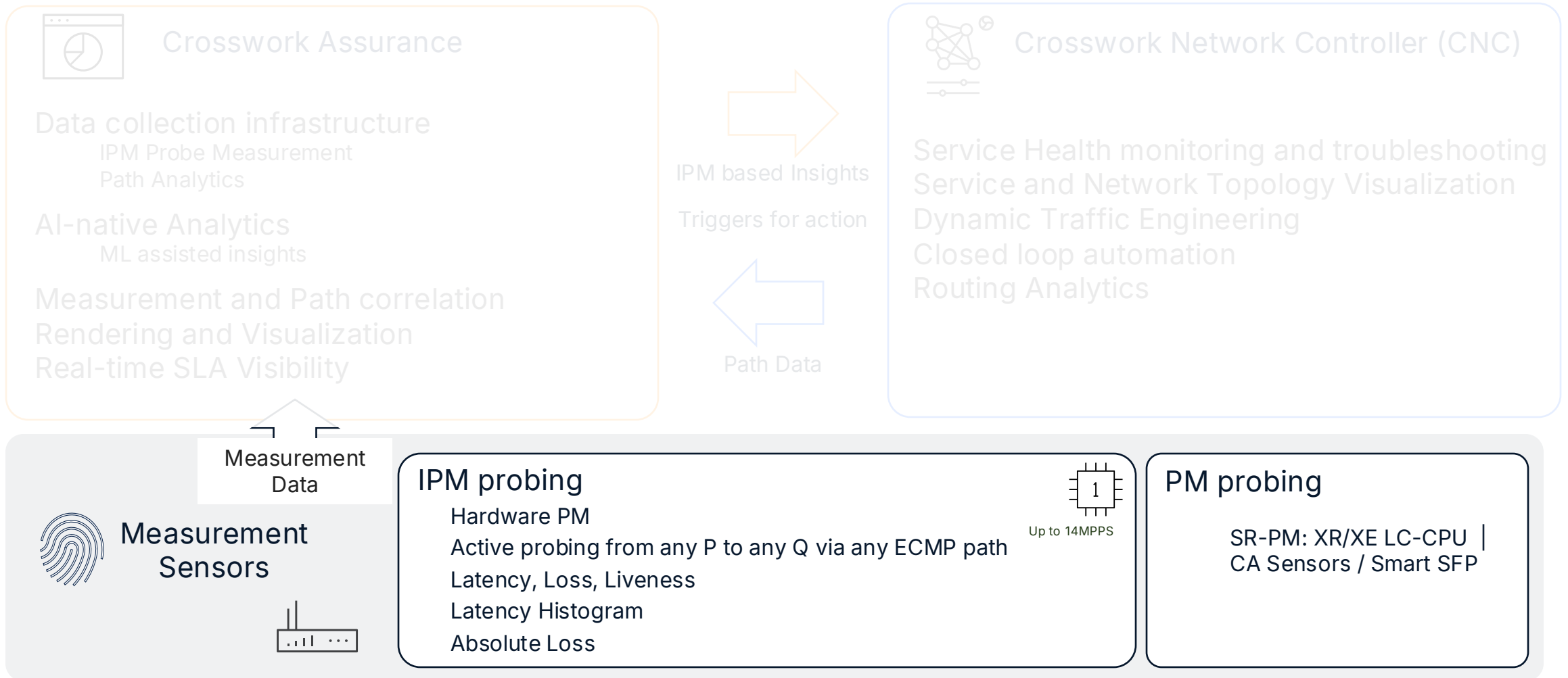


How?

Integrated Performance Measurement



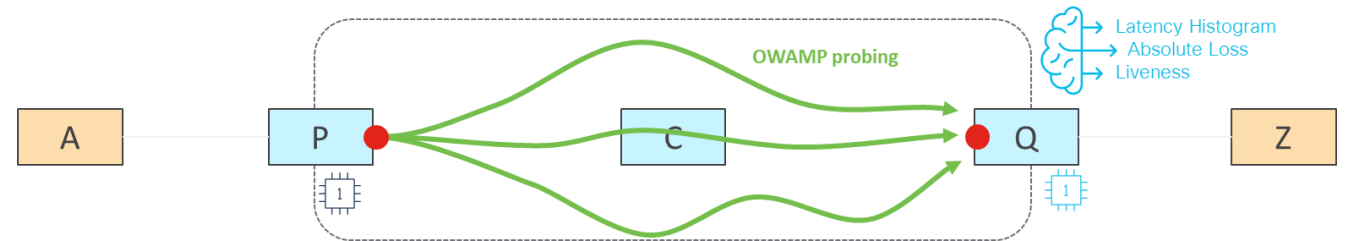
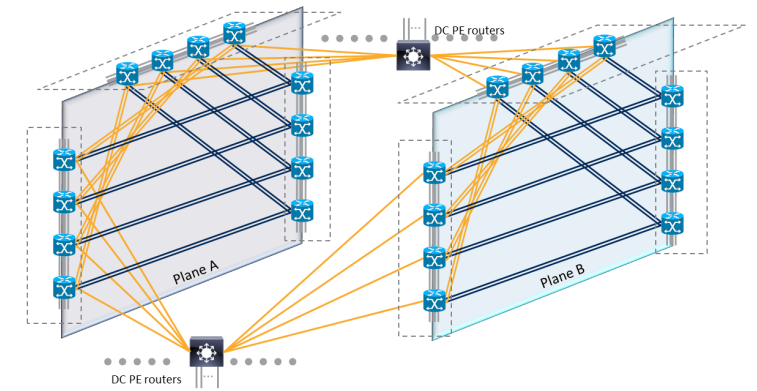
IP Measurement: Probing built into the fabric – 1 of 3



IP Measurement: Probing built into the fabric

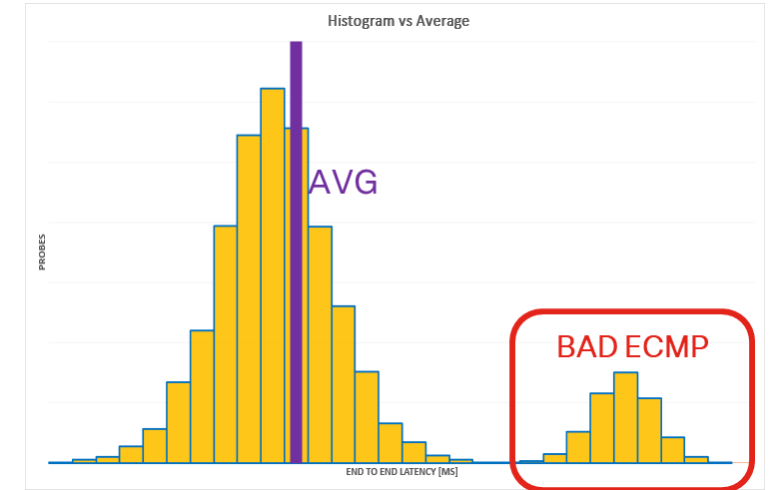
Innovation started from HW

- **Millions** of one-way probing packets out and in
- **Each probe measures** latency, loss and liveness
 - Latency **histogram** instead of min, avg, max
 - Reports the experience of the whole population
 - Digitalize latency curve with 28-bin histogram
- **Absolute loss** instead of loss approximations
 - Alternate marking (RFC9341)
- **Liveness detection**



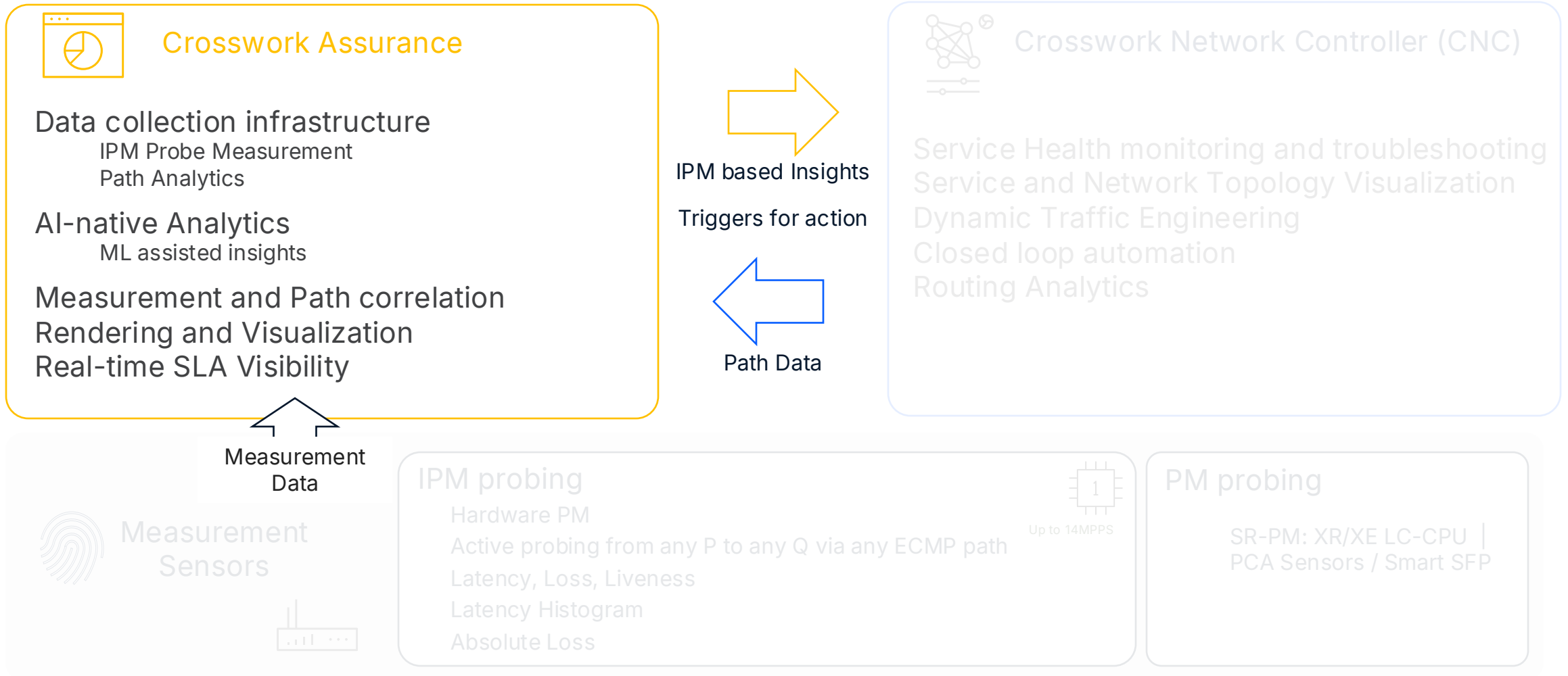
Accurate, Rich Metrics & Probing Flexibility

- Each probe measures 3L: **latency, loss and liveness**
- **Latency histogram** instead of min, avg, max
 - Reports the experience of the whole population
 - Digitalize latency curve with 28-bin histogram
- **Absolute loss** instead of loss approximations
 - Alternate marking (RFC9341)
- **Liveness detection**
- **Configurable** source, dest address and VRF, DSCP values
- **ECMP – 2 Modes of operation:**
 - Spray: every probe packet uses different FL values. Measure and aggregate across path
 - Dedicated-FL: each session uses a dedicated FL value for all probe packets
- Can combine with **uSID policy**
- **Stream all of the data with telemetry at Q**

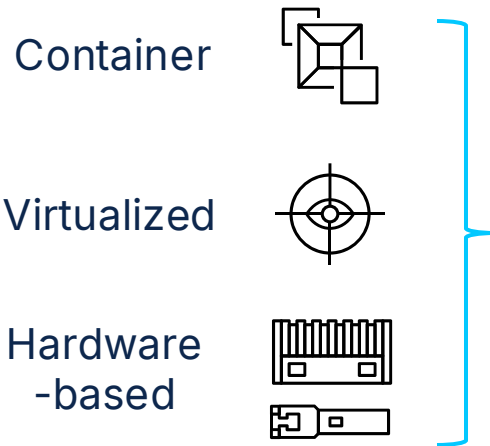
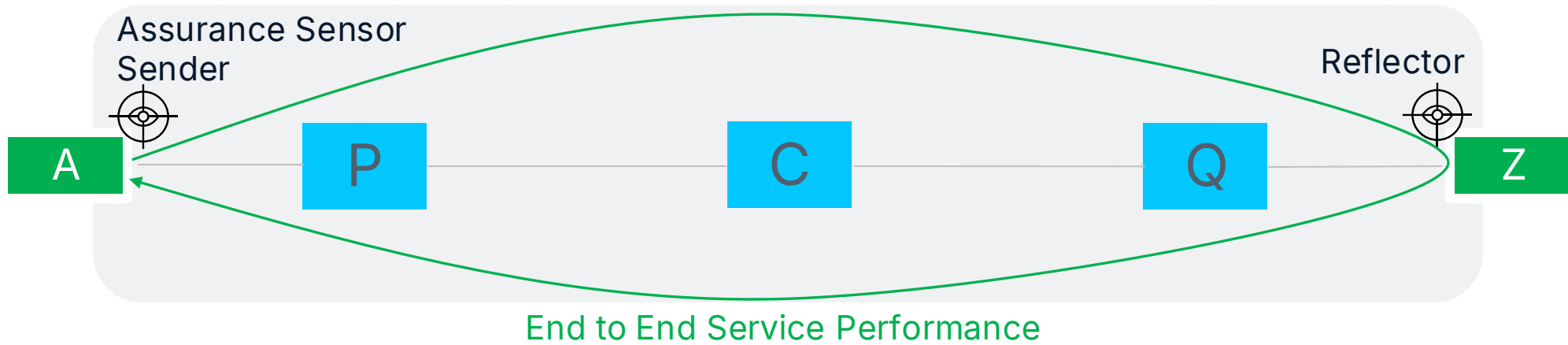


- 1 bad path out of 16 ECMP
- 6.25% of the clients impacted
- Average hides the issue
- Latency Histograms reports the experience of the whole population

Power of Analytics



Real-time SLA Visibility with Assurance Sensors



Fully configurable assurance sessions, e.g.

- VLAN 100
- 10s – 1000s of packets per second
- DSCP 46
- Result collection down to every second

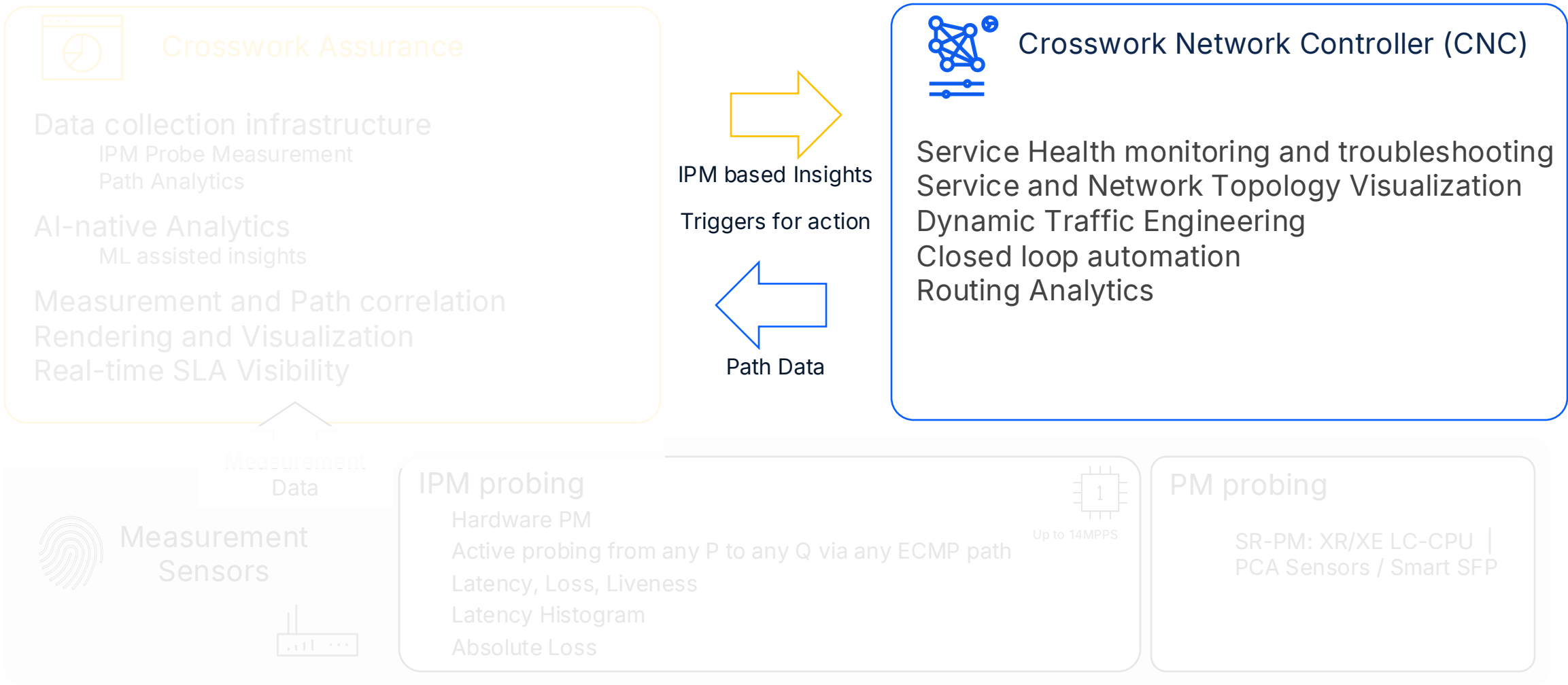
- Standards based (TWAMP RFC5357, Y.1731)
- Millisecond sampling (high packets per second rate)
- Microsecond accuracy for time-based measurements
- One-way directional metrics without use of external sync
- 50+ KPIs – Beyond the standard TWAMP KPIs (percentiles, etc.)

Understand Network Performance Every P to Q



- Network-wide view of all P-Q paths in the network with summary information over historical period.
- Select a specific P-Q path and drill down to see details.

Routing Analytics – 3 of 3



Routing Analytics defines the baseline

- RA provides a baseline for the measurements between each end-point pairs:
 - Is the measured latency good or bad ?
 - Reports the end-to-end **expected propagation delay** for each ECMP Path.

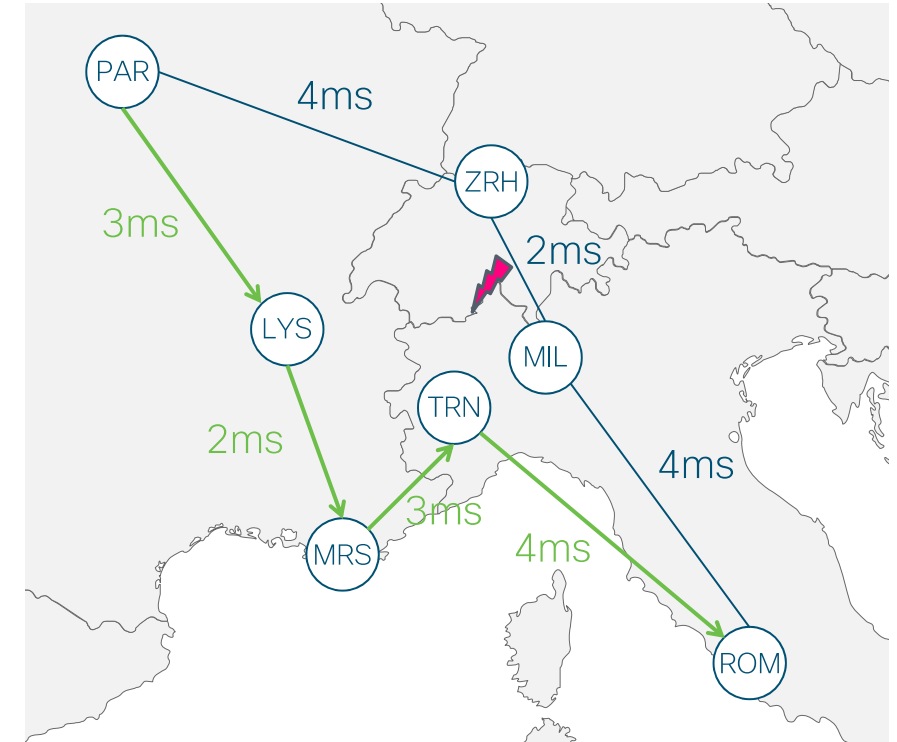


- Path from PAR to ROM on current topology:

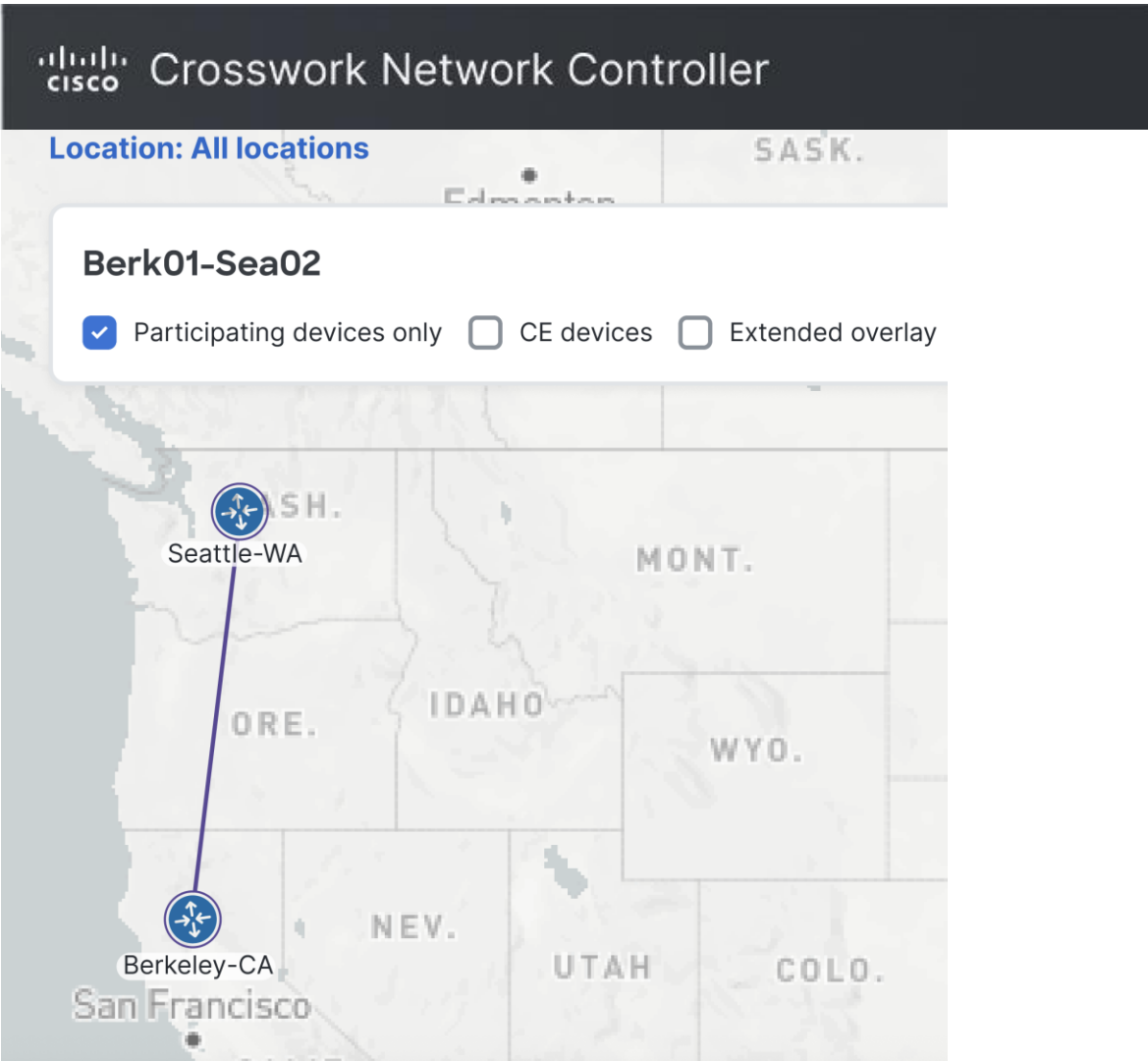
PAR – LYS – MRS – TRN – ROM

- End-to-end propagation delay:

$$3 + 2 + 3 + 4 = 12ms$$



CNC Provisioning IPM Probes



Service details

Provision IPM probes

Name	Berk01-Sea02	Health status	Degraded ⓘ
Provisioning	Success	Monitoring profile	L3VPN-1-GOLD ⓘ
Monitoring status	Error		

- Sender IPv6
- Receiver IPv6
- Latency Histogram Template

CNC Auto Populates Session ID
Provider Metadata to PCA



IPM in Action



Path Preview

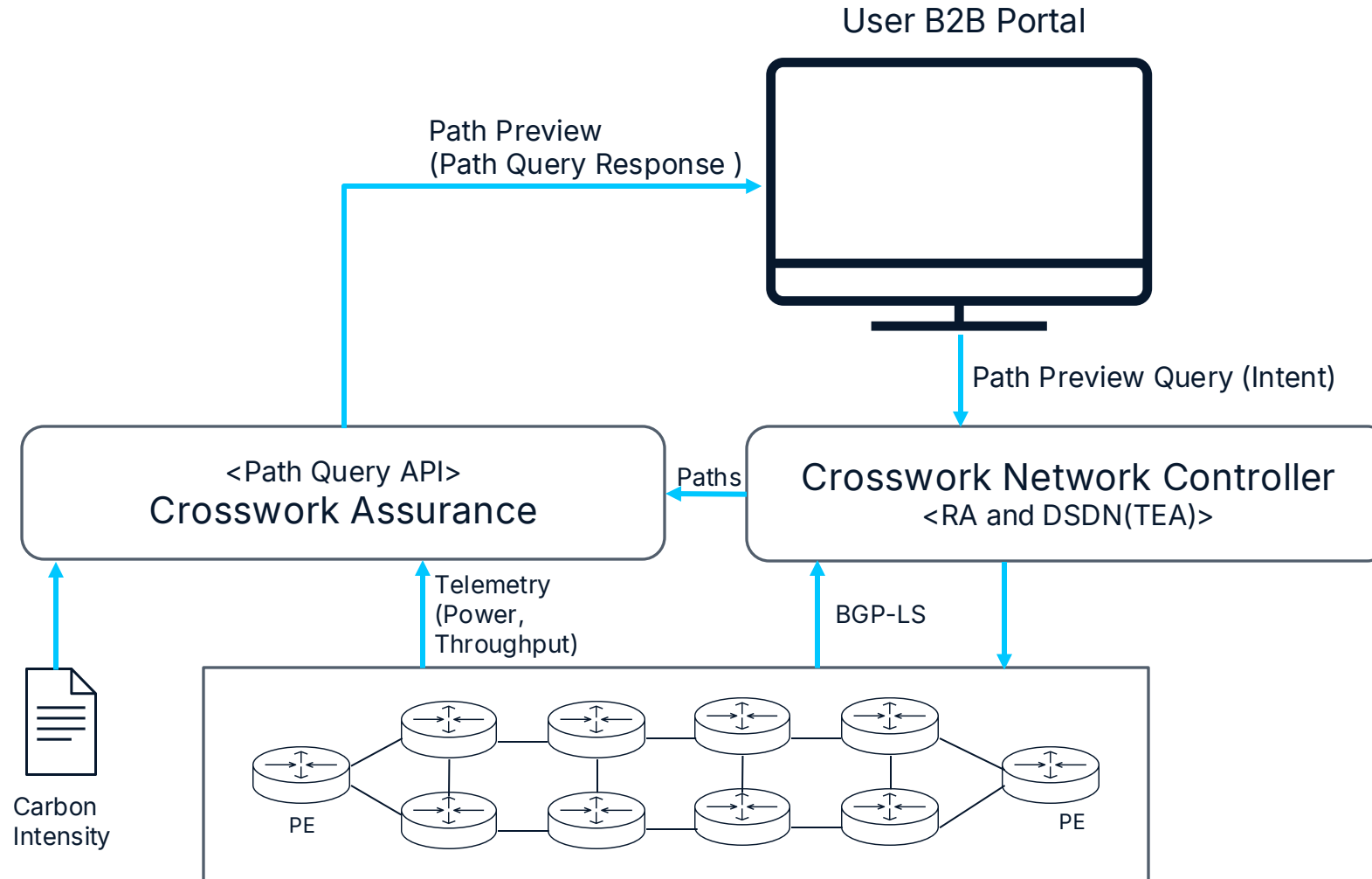


Network Latency Grid



Topology Aware Assurance

UC-1 NaaS: Path Preview





Demo

Good evening, Amish

Welcome to Cisco Cloud Control



Ask anything, / for prompts, # for knowledge

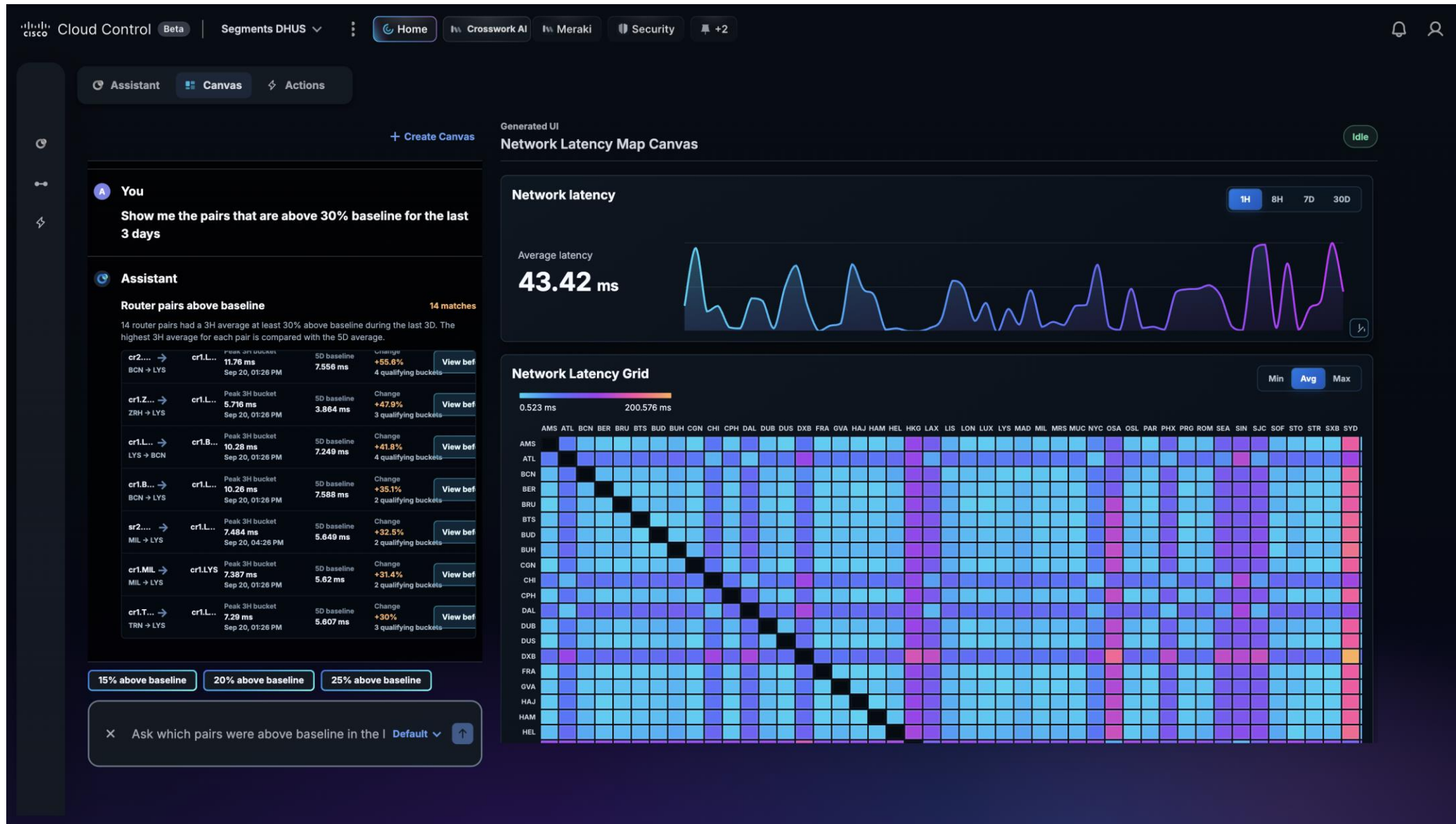
Default



Path Preview

Network Latency Map

UC-2: Network Latency Grid



UC-3: Topology Aware Assurance



Thank you



